

Table S2. Inter-season design decisions

Transition	Trigger / observed outcome	Decision	Rationale
S4 -> S5	Best S4 results hit SL/TP ceiling (8 %); threshold < 0.80 was a dead zone	Expand $\text{SL_RANGE}/\text{TP_RANGE}$ to 10 %; enforce $\text{THRESHOLD_RANGE} \geq 0.80$	Give Optuna room to explore wider exits and avoid low-threshold overtrading
S5 -> S9	Earlier seasons (S2–S8) had OOS contamination: Optuna touched 2025/2026	Introduce passive holdout: $\text{validation_years}=[2023, 2024]$, $\text{holdout_years}=[2020-2022]$ Optuna never sees holdout	Restore holdout integrity; modern
S9 -> S10	S9 validation [2023,2024] were both bull markets; best S9 holdout was negative	Add 2022 bear market to validation; introduce ATR kill-switch at $3 \times$ normal volatility	Force robustness to bear regimes and volatility spikes
S10 -> S11	S10 ratchet baseline caused progressive val-overfitting (holdout degraded)	Replace ratchet with triple gate: $\text{AUC} \geq 0.55$, $\text{Sharpe}(\text{val}) \geq 1$, $\text{Sharpe}(\text{holdout}) \geq 0.5$; no ratchet	Stop holdout degradation while retaining selection pressure
S11 -> S12	S11 succeeded on BNB; test cross-asset generalisation	Switch to BTC/USDT; train 2017–2020, val 2021–2023, holdout 2024–2025; relax holdout gate to 0.3	Verify asset-agnostic design on a different volatility regime

Note: Decisions were made by the human reviewer after each season based on the distribution of accepted strategies, failure modes, and holdout behaviour. Each transition typically required 15–30 minutes of review.